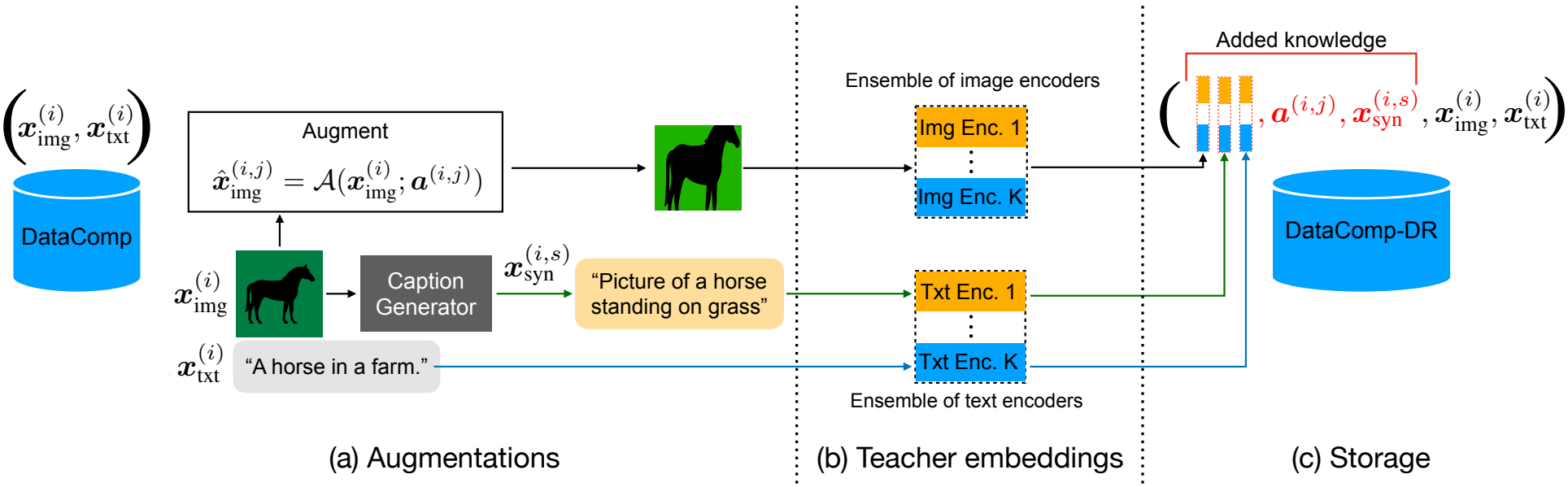
$(\boldsymbol{x}_{\text{img}}^{(i)}, \boldsymbol{x}_{\text{txt}}^{(i)})$
DataComp
Augment
$\hat{\boldsymbol{x}}_{\text{img}}^{(i,j)} = \mathcal{A}(\boldsymbol{x}_{\text{img}}^{(i)}; \boldsymbol{a}^{(i,j)})$
$\boldsymbol{x}_{\text{img}}^{(i)}$
Caption Generator
$\boldsymbol{x}_{\text{syn}}^{(i,s)}$
"Picture of a horse standing on grass"
$\boldsymbol{x}_{\text{txt}}^{(i)}$
"A horse in a farm."
Ensemble of image encoders
Img Enc. 1
Img Enc. K
Txt Enc. 1
Txt Enc. K
Ensemble of text encoders
Added knowledge
$(\boldsymbol{a}^{(i,j)}, \boldsymbol{x}_{\text{syn}}^{(i,s)}, \boldsymbol{x}_{\text{img}}^{(i)}, \boldsymbol{x}_{\text{txt}}^{(i)})$
DataComp-DR
(a) Augmentations
(b) Teacher embeddings
(c) Storage